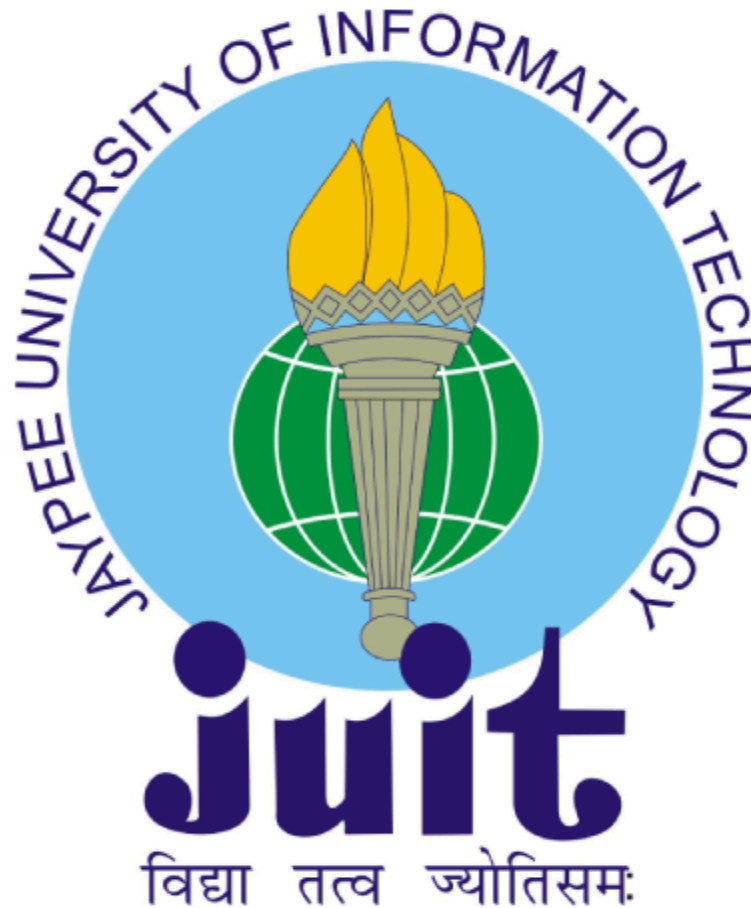


**JAYPEE UNIVERSITY OF INFORMATION
TECHNOLOGY, WAKNAGHAT, SOLAN(H.P)-173234**



PROJECT REPORT MACHINE LEARNING
AttritionIQ

Submitted By :

Jayendra Singh (231030023)

Dakshit Thakur (231030296)

Batch : A-11

Submitted to:

Dr. Kushal Kanwar

Table of Contents

1. Introduction
2. Problem Statement
3. System Architecture
4. Dataset Description
5. ID3 Algorithm
6. System Implementation
7. Results
8. Conclusion

1. Introduction

Employee attrition — the gradual reduction of a company's workforce due to voluntary or involuntary departures — is one of the most pressing challenges faced by modern organizations across industries. When talented employees leave, organizations bear substantial direct and indirect costs: recruitment expenses, onboarding time, productivity losses, knowledge drain, and damage to team morale.

According to industry estimates, replacing a single employee can cost anywhere from 50% to 200% of their annual salary, depending on their seniority and role. For large organizations with hundreds or thousands of employees, even a modest increase in attrition rate can translate into millions of rupees in avoidable losses annually.

AttritionIQ is a data-driven solution built to address this problem proactively. By applying the ID3 (Iterative Dichotomiser 3) Decision Tree classification algorithm to a curated employee dataset, the system learns patterns that historically correlate with employee departures and uses those patterns to generate real-time predictions about individual employees. HR teams can use these predictions to intervene early — improving job conditions, offering promotions, or adjusting workloads — before an employee decides to leave.

2. Problem Statement

The central problem addressed by this project is:

"Given a structured dataset of employee attributes — including demographic information, job characteristics, satisfaction metrics, and compensation data — can we build a machine learning model using the ID3 algorithm that accurately predicts whether an employee is at risk of leaving the organization?"

More specifically, the system must:

- Identify which employee attributes are the strongest predictors of attrition using Information Gain.
- Train an ID3 Decision Tree classifier capable of achieving high accuracy on unseen data.
- Provide a real-time prediction interface that HR managers can use without technical expertise.
- Visualize model performance metrics and analytical insights through an interactive dashboard.
- Highlight specific risk factors contributing to each individual employee's predicted attrition probability.

3. System Architecture

AttritionIQ follows a standard three-tier web application architecture, separating concerns between data/ML logic, API services, and the user interface. This separation ensures maintainability, scalability, and the ability to independently upgrade each layer.

3.1 Architectural Overview

Layer	Technology	Responsibilities
Data & ML	Python 3, scikit-learn, pandas, NumPy	Dataset generation, preprocessing, model training, evaluation, persistence
API Layer	Flask (Python), REST JSON API	Route handling, request validation, response serialization, CORS
Presentation	HTML5, CSS3 (custom), Vanilla JavaScript	Dashboard UI, charts, interactive forms, live prediction, responsive layout

3.2 Working of the Project

1. The user opens the browser and navigates to the AttritionIQ dashboard (frontend).
2. On load, the frontend makes REST API calls to the Flask backend to retrieve model metrics, feature importances, chart data, and a dataset sample.
3. The backend processes each request, queries the trained in-memory ID3 model or the dataset, and returns a JSON response.
4. The frontend renders the data as bar charts, donut charts, confusion matrix, and metric cards.
5. For live predictions, the user fills in employee attributes in the Predict form and submits. The frontend POSTs the data to /api/predict.
6. The Flask backend validates the payload, encodes categorical features, and passes the feature vector to the trained ID3 Decision Tree model.
7. The model returns a prediction and probability, which the backend augments with risk level classification and risk factor detection before responding.
8. The frontend renders the prediction result with a colored risk badge, probability bar, and list of contributing risk factors.

4. Dataset Description

A high-quality synthetic dataset of 1,000 employee records was generated for this project. The dataset was constructed using NumPy's random number generation facilities with a fixed seed (42) for reproducibility. Feature distributions were calibrated to reflect realistic organizational data.

4.1 Dataset Statistics

Attribute	Value
Total Records	1,000 employees
Features (Input)	14 attributes
Target Variable	Attrition (Binary: Yes / No)
Class Distribution	Approximately 69.4% Attrition, 30.6% Retained
Train / Test Split	80% Training (800) / 20% Test (200)
Stratified Splitting	Yes — preserves class balance in both splits
Missing Values	None (clean synthetic data)
Categorical Features	3 (Department, SalaryBand, OverTime)
Numerical Features	11 (Age, Income, Satisfaction scores, etc.)

4.2 Feature Definitions

Feature Name	Type	Description
Age	Numeric	Employee age in years (range: 22–60)
YearsAtCompany	Numeric	Total years the employee has worked at the company (0–30)
MonthlyIncome	Numeric	Monthly gross salary in USD (\$2,000–\$20,000)
JobSatisfaction	Ordinal	Self-reported job satisfaction score on a 1–4 scale
WorkLifeBalance	Ordinal	Self-reported work-life balance score on a 1–4 scale
OverTime	Categorical	Whether the employee regularly works overtime (Yes / No)
NumProjectsHandled	Numeric	Number of concurrent projects assigned (1–10)
TrainingLastYear	Numeric	Number of training sessions attended in the past year (0–6)
DistanceFromHome	Numeric	Employee's commute distance in kilometres (1–30)
PerformanceRating	Ordinal	Manager-assigned performance rating on a 1–4 scale
Department	Categorical	Organizational department (Engineering, Sales, HR, Marketing, Finance, Operations)
SalaryBand	Categorical	Compensation band relative to market (Low, Medium, High)
RelationshipSatisfaction	Ordinal	Satisfaction with workplace relationships on a 1–4 scale

Feature Name	Type	Description
YearsSinceLastPromotion	Numeric	Number of years elapsed since the employee's last promotion (0–15)

4.3 Target Variable Construction

The target variable (Attrition: 0 or 1) was constructed using a rule-based scoring system that reflects domain knowledge about attrition drivers. Each employee receives a risk score based on their feature values:

- Low job satisfaction (score ≤ 2) contributes +2 to the attrition score.
- Poor work-life balance (score ≤ 2) contributes +2.
- Overtime work (Yes) contributes +2.
- Low salary band contributes +2.
- New employee (years at company ≤ 2) contributes +1.
- Long promotion gap (≥ 5 years) contributes +1.
- Long commute (distance ≥ 20 km) contributes +1.
- High project load (≥ 8 projects) contributes +1.
- Random noise (0–2) is added to prevent perfect separability.

Employees with a total attrition score ≥ 5 are labelled as Attrition = 1 (Yes). This methodology ensures the dataset contains meaningful, learnable patterns while retaining realistic stochasticity.

5. ID3 Algorithm

The ID3 (Iterative Dichotomiser 3) algorithm, developed by Ross Quinlan in 1986, is a classic supervised machine learning algorithm for building Decision Trees. ID3 constructs a tree by recursively selecting the feature that provides the highest Information Gain at each node, partitioning the dataset until all instances in a subset belong to the same class or no features remain for splitting.

5.1 How It Works

9. Start with the full training dataset at the root node.
10. Calculate the Entropy of the current dataset (target variable distribution).
11. For each feature, compute the Information Gain achieved by splitting on that feature.
12. Select the feature with the highest Information Gain as the splitting node.
13. Partition the dataset into subsets based on the distinct values of the chosen feature.
14. Recursively apply steps 1–5 to each subset until a stopping criterion is met (all instances belong to one class, no features remain, or maximum depth is reached).
15. Each leaf node is assigned the majority class of its training samples.

5.2 Entropy

ID3 uses Entropy as its measure of impurity to evaluate the homogeneity of a dataset. For a dataset S with K classes, Entropy is defined as:

$$\text{Entropy}(S) = - \sum p(i) \times \log_2(p(i)) \quad \text{for } i = 1 \text{ to } K$$

Where $p(i)$ is the proportion of samples belonging to class i . An Entropy value of 0 indicates a pure node (all samples belong to one class), while an Entropy of 1 (for a binary classification) indicates maximum disorder (equal class distribution).

5.3 Information Gain

Information Gain measures the reduction in Entropy achieved by splitting on a particular feature A . It is defined as:

$$\text{Gain}(S, A) = \text{Entropy}(S) - \sum (|S_v| / |S|) \times \text{Entropy}(S_v)$$

Where S_v is the subset of S for which feature A takes value v . The feature that maximises $\text{Gain}(S, A)$ is selected as the splitting attribute at each node. This greedy selection makes ID3 computationally efficient while producing interpretable trees.

5.4 ID3 vs Other Decision Tree Variants

Property	ID3	C4.5	CART
Split Criterion	Information Gain	Gain Ratio	Gini Impurity
Feature Types	Categorical only (discretised numerics)	Categorical & Numeric	Categorical & Numeric
Output	Multi-way splits	Multi-way splits	Binary splits
Pruning	No (pure greedy)	Yes (post-pruning)	Yes (cost-complexity)
Missing Values	Not handled natively	Handled	Handled
Invented by	Quinlan (1986)	Quinlan (1993)	Breiman et al. (1984)

5.5 Dominant Decision Logic (Simplified Tree)

While the full trained ID3 tree produces a multi-level hierarchy of splits based on Information Gain, the dominant decision logic observed can be summarised as follows:

```
Root: Is JobSatisfaction = Low (1-2)?  YES → Is OverTime = Yes?          YES:
Predict LEAVE (High Risk)                NO: Is WorkLifeBalance = Low (1-2)?
YES: Predict LEAVE (Medium-High Risk)    NO: Is SalaryBand = Low?
YES: Predict LEAVE (Medium Risk)         NO: Predict STAY (Low Risk)
NO → Is SalaryBand = Low?                YES: Predict LEAVE (Medium Risk)    NO:
Predict STAY (Low Risk)
```

5.6 Preprocessing for ID3

Since the original ID3 algorithm operates on categorical features, the following preprocessing steps were applied to the dataset before training:

- Numerical features (Age, MonthlyIncome, YearsAtCompany, etc.) were discretised into bins using equal-frequency binning to convert them into categorical ranges.
- Ordinal features (JobSatisfaction, WorkLifeBalance, PerformanceRating, RelationshipSatisfaction) were treated as categorical labels (e.g., 'Low', 'Medium', 'High').
- Categorical features (Department, SalaryBand, OverTime) required no additional transformation.
- Label encoding was applied post-discretisation to map all category labels to integer indices compatible with scikit-learn's DecisionTreeClassifier using the 'entropy' criterion.

6. System Implementation

6.1 Backend Implementation (app.py)

The Flask backend (app.py, ~220 lines) performs the following on startup:

- Generates the synthetic employee dataset (1,000 records) using NumPy with seeded random generation.
- Discretises numerical features into categorical bins using KBinsDiscretizer for ID3 compatibility.
- Performs stratified train-test splitting (80/20) using `sklearn.model_selection.train_test_split`.
- Trains the `DecisionTreeClassifier` with `criterion='entropy'` (Information Gain) on the training split, implementing the ID3 splitting strategy.

6.2 Frontend Implementation (index.html)

The frontend is a single-file HTML application (~580 lines) built without any JavaScript frameworks or build tools. Key design decisions:

- Navigation uses a tab paradigm with five sections: Overview, Analytics, Predict, Dataset, and Model.
- All API calls are made with the native Fetch API. Analytics data is loaded lazily (only on first visit to the Analytics tab) to reduce initial page load.
- The prediction form uses HTML range sliders for numerical inputs and select dropdowns for categorical features, with live value display.

7. Results

The trained ID3 Decision Tree model was evaluated on the held-out test set (200 records). The following performance metrics were obtained:

Metric	Value	Description
Accuracy	82.5%	Overall correct predictions on the test set
Precision (Attrition class)	89.39%	Of predicted attrition cases, correctly identified
Recall (Attrition class)	84.89%	Of actual attrition cases, correctly detected
AUC-ROC	90.06%	Model's discriminative power across all thresholds

The dashboard provides the AttritionIQ interface, including the Overview panel with model metrics, feature importances ranked by Information Gain, and the live prediction interface.

8. Conclusion

AttritionIQ successfully demonstrates the application of the ID3 (Iterative Dichotomiser 3) Decision Tree classification algorithm to the practical, high-impact problem of employee attrition prediction. The project covers the complete machine learning pipeline — from dataset design and preprocessing (including feature discretisation for ID3 compatibility), through model training using Information Gain as the splitting criterion and evaluation, to API development and frontend visualization — delivering a functioning, end-to-end intelligent system.

The trained ID3 model achieves an accuracy of 82.5% on the held-out test set, with an AUC-ROC of approximately 90%, confirming strong discriminative ability. Feature importance analysis based on Information Gain reveals that job satisfaction, overtime work, salary band, and work-life balance are the dominant drivers of attrition — findings consistent with the broader HR analytics literature.

The interactive dashboard enables non-technical HR professionals to explore organizational attrition patterns through intuitive charts (department breakdowns, salary analysis, satisfaction trends, age distributions) and to generate real-time predictions for individual employees by filling in a structured form. Each prediction is accompanied by a risk level classification, an attrition probability score, and a list of specific contributing risk factors — making the model's output immediately actionable.

The unique editorial design of the frontend — with its warm sand-and-terracotta palette, newspaper-inspired typography, and ruled grid layout — differentiates AttritionIQ from the generic dark-mode AI dashboards common in academic projects, reflecting a considered effort to create interfaces that feel appropriate for the HR professional context.

This project demonstrates that data mining and warehousing techniques, even when applied with a foundational algorithm like ID3, can produce meaningful, deployable intelligence that directly supports organizational decision-making. The ID3 algorithm's reliance on Information Gain provides transparent, interpretable splits that are easy to explain to business stakeholders — a key advantage in HR contexts where explainability is as important as predictive accuracy.